

15 Digital camera interface (DCMI)

This section applies to all STM32F4xx devices, unless otherwise specified.

15.1 DCMI introduction

The digital camera is a synchronous parallel interface able to receive a high-speed data flow from an external 8-, 10-, 12- or 14-bit CMOS camera module. It supports different data formats: YCbCr4:2:2/RGB565 progressive video and compressed data (JPEG).

This interface is for use with black & white cameras, X24 and X5 cameras, and it is assumed that all pre-processing like resizing is performed in the camera module.

15.2 DCMI main features

- 8-, 10-, 12- or 14-bit parallel interface
- Embedded/external line and frame synchronization
- Continuous or snapshot mode
- Crop feature
- Supports the following data formats:
 - 8/10/12/14- bit progressive video: either monochrome or raw bayer
 - YCbCr 4:2:2 progressive video
 - RGB 565 progressive video
 - Compressed data: JPEG

15.3 DCMI pins

[Table 77](#) shows the DCMI pins.

Table 77. DCMI pins

Name	Signal type
D[0:13]	Data inputs
HSYNC	Horizontal synchronization input
VSYNC	Vertical synchronization input
PIXCLK	Pixel clock input

15.4 DCMI clocks

The digital camera interface uses two clock domains PIXCLK and HCLK. The signals generated with PIXCLK are sampled on the rising edge of HCLK once they are stable. An enable signal is generated in the HCLK domain, to indicate that data coming from the camera are stable and can be sampled. The minimum PIXCLK period must be higher than 2.5 HCLK periods.

15.5 DCMI functional overview

The digital camera interface is a synchronous parallel interface that can receive high-speed (up to 54 Mbytes/s) data flows. It consists of up to 14 data lines (D13-D0) and a pixel clock line (PIXCLK). The pixel clock has a programmable polarity, so that data can be captured on either the rising or the falling edge of the pixel clock.

The data are packed into a 32-bit data register (DCMI_DR) and then transferred through a general-purpose DMA channel. The image buffer is managed by the DMA, not by the camera interface.

The data received from the camera can be organized in lines/frames (raw YUB/RGB/Bayer modes) or can be a sequence of JPEG images. To enable JPEG image reception, the JPEG bit (bit 3 of DCMI_CR register) must be set.

The data flow is synchronized either by hardware using the optional HSYNC (horizontal synchronization) and VSYNC (vertical synchronization) signals or by synchronization codes embedded in the data flow.

Figure 72 shows the DCMI block diagram.

Figure 72. DCMI block diagram

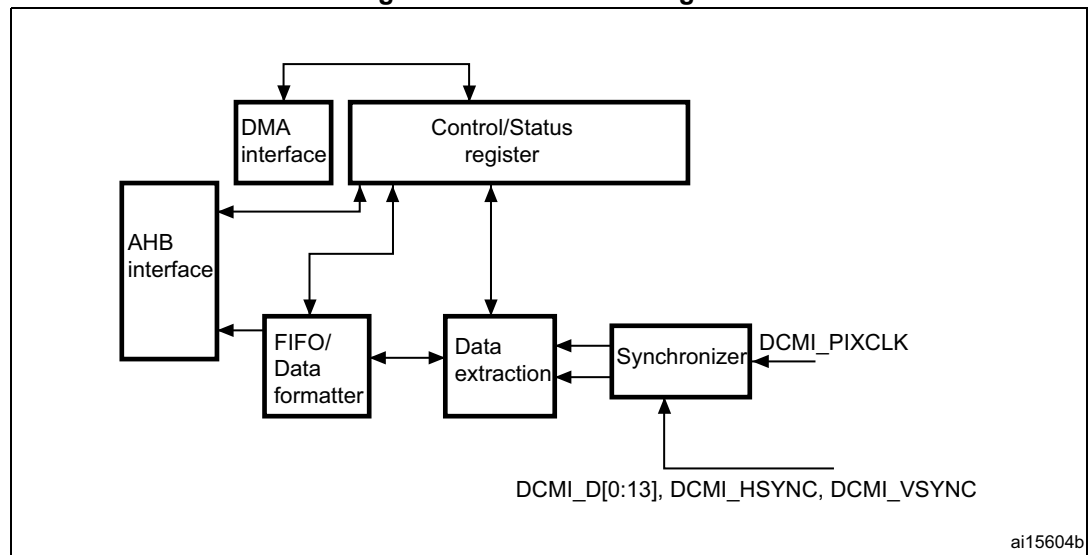
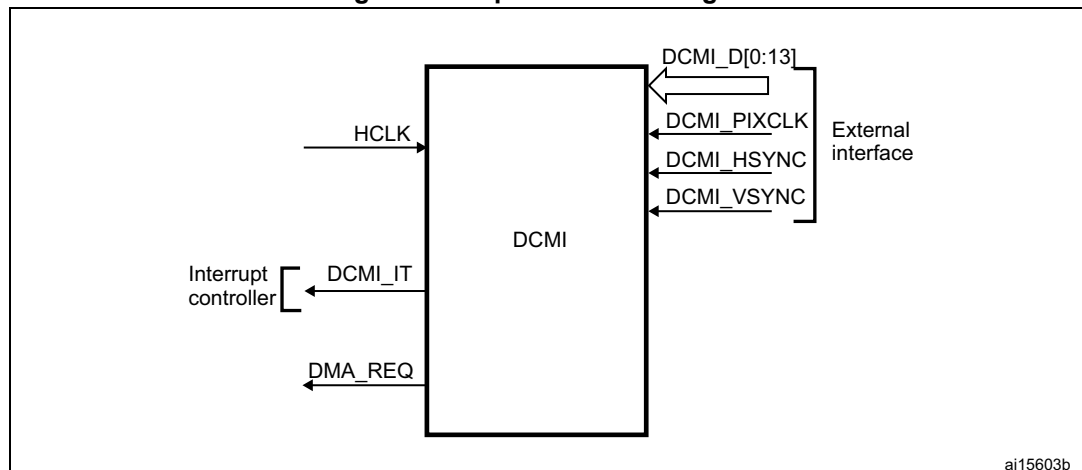


Figure 73. Top-level block diagram



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15.5.1 DMA interface

The DMA interface is active when the CAPTURE bit in the DCMI_CR register is set. A DMA request is generated each time the camera interface receives a complete 32-bit data block in its register.

15.5.2 DCMI physical interface

The interface is composed of 11/13/15/17 inputs. Only the Slave mode is supported.

The camera interface can capture 8-bit, 10-bit, 12-bit or 14-bit data depending on the EDM[1:0] bits in the DCMI_CR register. If less than 14 bits are used, the unused data pins must not be assigned to DCMI interface through GPIO alternate functions.

Table 78. DCMI signals

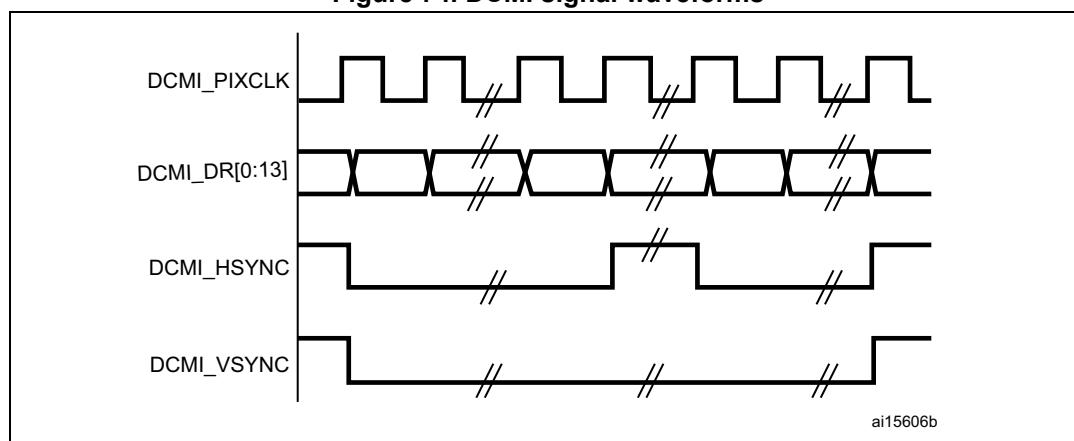
Signal name		Signal description
8 bits	D[0..7]	Data
10 bits	D[0..9]	
12 bits	D[0..11]	
14 bits	D[0..13]	
PIXCLK		Pixel clock
HSYNC		Horizontal synchronization / Data valid
VSYNC		Vertical synchronization

The data are synchronous with PIXCLK and change on the rising/falling edge of the pixel clock depending on the polarity.

The HSYNC signal indicates the start/end of a line.

The VSYNC signal indicates the start/end of a frame

Figure 74. DCMI signal waveforms



1. The capture edge of DCMI_PIXCLK is the falling edge, the active state of DCMI_HSYNC and DCMI_VSYNC is 1.
1. DCMI_HSYNC and DCMI_VSYNC can change states at the same time.

8-bit data

When EDM[1:0] in DCMI_CR are programmed to “00” the interface captures 8 LSB’s at its input (D[0:7]) and stores them as 8-bit data. The D[13:8] inputs are ignored. In this case, to capture a 32-bit word, the camera interface takes four pixel clock cycles.

The first captured data byte is placed in the LSB position in the 32-bit word and the 4th captured data byte is placed in the MSB position in the 32-bit word. [Table 79](#) gives an example of the positioning of captured data bytes in two 32-bit words.

Table 79. Positioning of captured data bytes in 32-bit words (8-bit width)

Byte address	31:24	23:16	15:8	7:0
0	D _{n+3} [7:0]	D _{n+2} [7:0]	D _{n+1} [7:0]	D _n [7:0]
4	D _{n+7} [7:0]	D _{n+6} [7:0]	D _{n+5} [7:0]	D _{n+4} [7:0]

10-bit data

When EDM[1:0] in DCMI_CR are programmed to “01”, the camera interface captures 10-bit data at its input D[0..9] and stores them as the 10 least significant bits of a 16-bit word. The remaining most significant bits in the DCMI_DR register (bits 11 to 15) are cleared to zero. So, in this case, a 32-bit data word is made up every two pixel clock cycles.

The first captured data are placed in the LSB position in the 32-bit word and the 2nd captured data are placed in the MSB position in the 32-bit word as shown in [Table 80](#).

Table 80. Positioning of captured data bytes in 32-bit words (10-bit width)

Byte address	31:26	25:16	15:10	9:0
0	0	D _{n+1} [9:0]	0	D _n [9:0]
4	0	D _{n+3} [9:0]	0	D _{n+2} [9:0]

12-bit data

When EDM[1:0] in DCMI_CR are programmed to “10”, the camera interface captures the 12-bit data at its input D[0..11] and stores them as the 12 least significant bits of a 16-bit word. The remaining most significant bits are cleared to zero. So, in this case a 32-bit data word is made up every two pixel clock cycles.

The first captured data are placed in the LSB position in the 32-bit word and the 2nd captured data are placed in the MSB position in the 32-bit word as shown in [Table 81](#).

Table 81. Positioning of captured data bytes in 32-bit words (12-bit width)

Byte address	31:28	27:16	15:12	11:0
0	0	D _{n+1} [11:0]	0	D _n [11:0]
4	0	D _{n+3} [11:0]	0	D _{n+2} [11:0]

14-bit data

When EDM[1:0] in DCMI_CR are programmed to “11”, the camera interface captures the 14-bit data at its input D[0..13] and stores them as the 14 least significant bits of a 16-bit word. The remaining most significant bits are cleared to zero. So, in this case a 32-bit data word is made up every two pixel clock cycles.

The first captured data are placed in the LSB position in the 32-bit word and the 2nd captured data are placed in the MSB position in the 32-bit word as shown in [Table 82](#).

Table 82. Positioning of captured data bytes in 32-bit words (14-bit width)

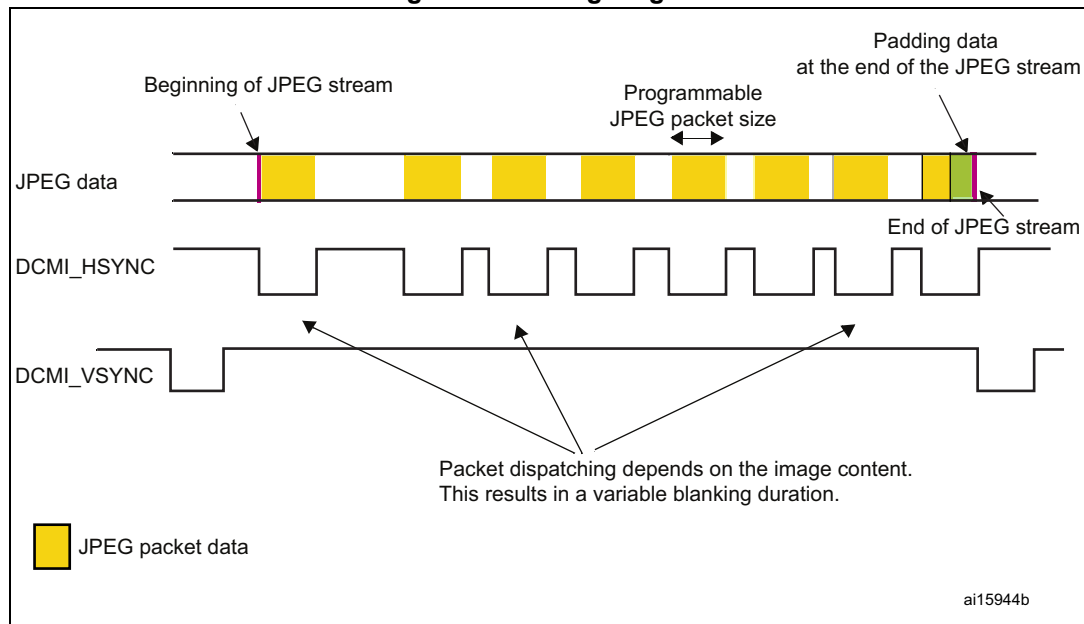
Byte address	31:30	29:16	15:14	13:0
0	0	D _{n+1} [13:0]	0	D _n [13:0]
4	0	D _{n+3} [13:0]	0	D _{n+2} [13:0]

15.5.3 Synchronization

The digital camera interface supports embedded or hardware (HSYNC & VSYNC) synchronization. When embedded synchronization is used, it is up to the digital camera module to make sure that the 0x00 and 0xFF values are used ONLY for synchronization (not in data). Embedded synchronization codes are supported only for the 8-bit parallel data interface width (that is, in the DCMI_CR register, the EDM[1:0] bits should be cleared to “00”).

For compressed data, the DCMI supports only the hardware synchronization mode. In this case, VSYNC is used as a start/end of the image, and HSYNC is used as a Data Valid signal. [Figure 75](#) shows the corresponding timing diagram.

Figure 75. Timing diagram



Hardware synchronization mode

In hardware synchronisation mode, the two synchronization signals (HSYNC/VSYNC) are used.

Depending on the camera module/mode, data may be transmitted during horizontal/vertical synchronisation periods. The HSYNC/VSYNC signals act like blanking signals since all the data received during HSYNC/VSYNC active periods are ignored.

In order to correctly transfer images into the DMA/RAM buffer, data transfer is synchronized with the VSYNC signal. When the hardware synchronisation mode is selected, and capture is enabled (CAPTURE bit set in DCMI_CR), data transfer is synchronized with the deactivation of the VSYNC signal (next start of frame).

Transfer can then be continuous, with successive frames transferred by DMA to successive buffers or the same/circular buffer. To allow the DMA management of successive frames, a VSIF (Vertical synchronization interrupt flag) is activated at the end of each frame.

Embedded data synchronization mode

In this synchronisation mode, the data flow is synchronised using 32-bit codes embedded in the data flow. These codes use the 0x00/0xFF values that are *not* used in data anymore. There are 4 types of codes, all with a 0xFF0000XY format. The embedded synchronization codes are supported only in 8-bit parallel data width capture (in the DCMI_CR register, the EDM[1:0] bits should be programmed to "00"). For other data widths, this mode generates unpredictable results and must not be used.

Note: Camera modules can have 8 such codes (in interleaved mode). For this reason, the interleaved mode is not supported by the camera interface (otherwise, every other half-frame would be discarded).

- Mode 2

Four embedded codes signal the following events

- Frame start (FS)
- Frame end (FE)
- Line start (LS)
- Line end (LE)

The XY values in the 0xFF0000XY format of the four codes are programmable (see [Section 15.8.7](#)).

A 0xFF value programmed as a “frame end” means that all the unused codes are interpreted as valid frame end codes.

In this mode, once the camera interface has been enabled, the frame capture starts after the first occurrence of the frame end (FE) code followed by a frame start (FS) code.

- Mode 1

An alternative coding is the camera mode 1. This mode is ITU656 compatible.

The codes signal another set of events:

- SAV (active line) - line start
- EAV (active line) - line end
- SAV (blanking) - end of line during interframe blanking period
- EAV (blanking) - end of line during interframe blanking period

This mode can be supported by programming the following codes:

- $FS \leq 0xFF$
- $FE \leq 0xFF$
- $LS \leq SAV \text{ (active)}$
- $LE \leq EAV \text{ (active)}$

An embedded unmask code is also implemented for frame/line start and frame/line end codes. Using it, it is possible to compare only the selected unmasked bits with the programmed code. You can therefore select a bit to compare in the embedded code and detect a frame/line start or frame/line end. This means that there can be different codes for the frame/line start and frame/line end with the unmasked bit position remaining the same.

Example

FS = 0xA5

Unmask code for FS = 0x10

In this case the frame start code is embedded in the bit 4 of the frame start code.

15.5.4 Capture modes

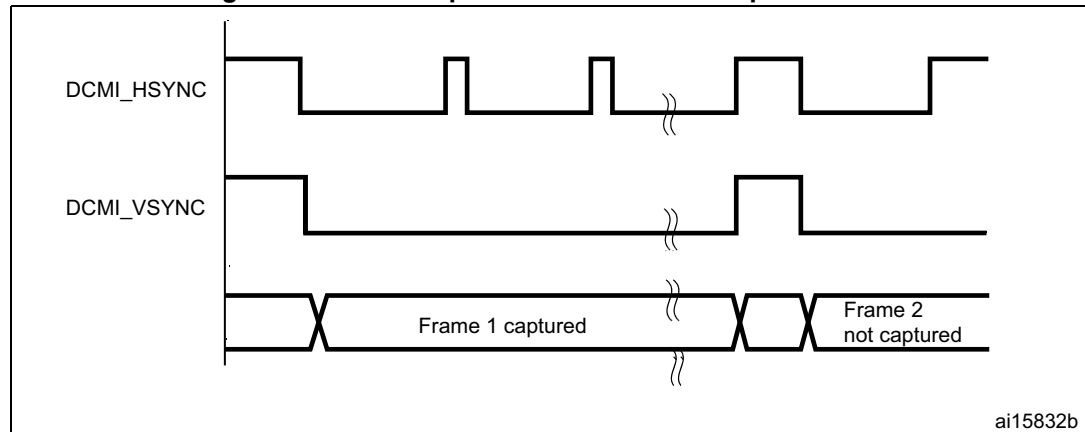
This interface supports two types of capture: snapshot (single frame) and continuous grab.

Snapshot mode (single frame)

In this mode, a single frame is captured (CM = '1' in the DCMI_CR register). After the CAPTURE bit is set in DCMI_CR, the interface waits for the detection of a start of frame before sampling the data. The camera interface is automatically disabled (CAPTURE bit cleared in DCMI_CR) after receiving the first complete frame. An interrupt is generated (IT_FRAME) if it is enabled.

In case of an overrun, the frame is lost and the CAPTURE bit is cleared.

Figure 76. Frame capture waveforms in Snapshot mode

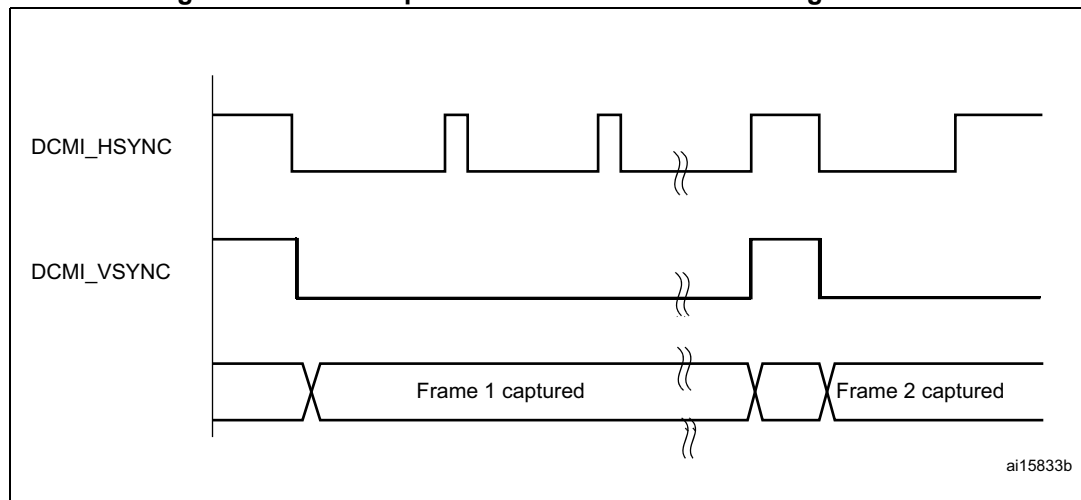


1. Here, the active state of DCMI_HSYNC and DCMI_VSYNC is 1.
2. DCMI_HSYNC and DCMI_VSYNC can change states at the same time.

Continuous grab mode

In this mode (CM bit = '0' in DCMI_CR), once the CAPTURE bit has been set in DCMI_CR, the grabbing process starts on the next VSYNC or embedded frame start depending on the mode. The process continues until the CAPTURE bit is cleared in DCMI_CR. Once the CAPTURE bit has been cleared, the grabbing process continues until the end of the current frame.

Figure 77. Frame capture waveforms in continuous grab mode



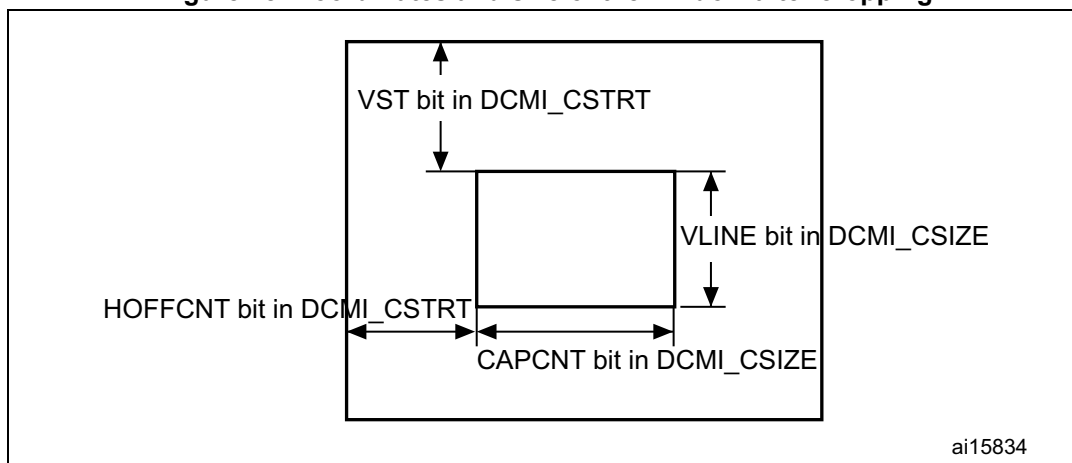
1. Here, the active state of DCMI_HSYNC and DCMI_VSYNC is 1.
2. DCMI_HSYNC and DCMI_VSYNC can change states at the same time.

In continuous grab mode, you can configure the FCRC bits in DCMI_CR to grab all pictures, every second picture or one out of four pictures to decrease the frame capture rate.

Note: *In the hardware synchronization mode (ESS = '0' in DCMI_CR), the IT_VSYNC interrupt is generated (if enabled) even when CAPTURE = '0' in DCMI_CR so, to reduce the frame capture rate even further, the IT_VSYNC interrupt can be used to count the number of frames between 2 captures in conjunction with the Snapshot mode. This is not allowed by embedded data synchronization mode.*

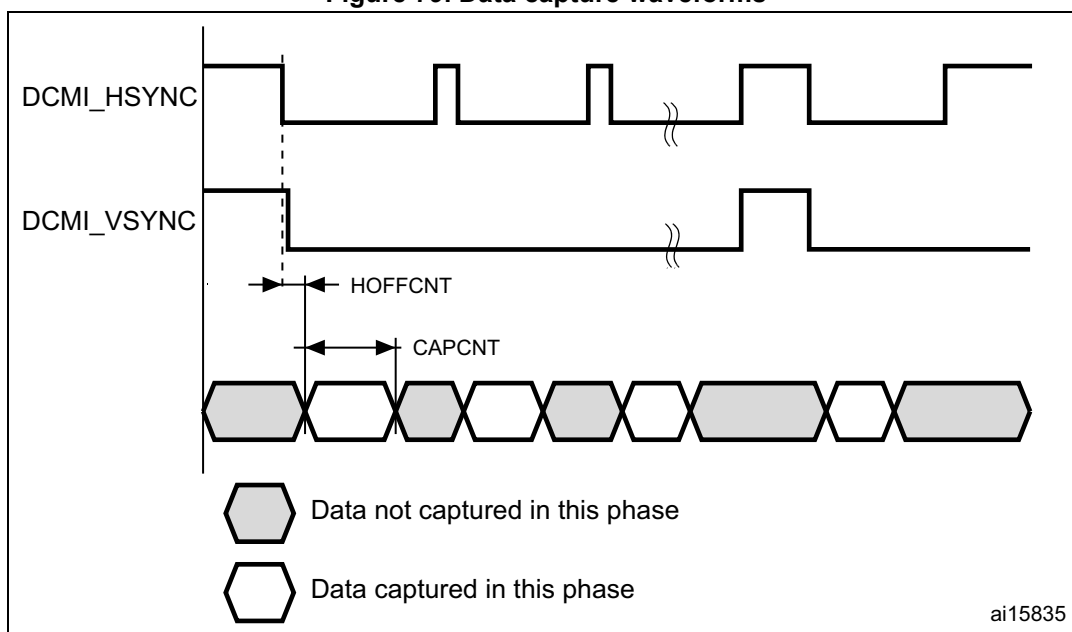
15.5.5 Crop feature

With the crop feature, the camera interface can select a rectangular window from the received image. The start (upper left corner) coordinates and size (horizontal dimension in number of pixel clocks and vertical dimension in number of lines) are specified using two 32-bit registers (DCMI_CWSTRT and DCMI_CWSIZE). The size of the window is specified in number of pixel clocks (horizontal dimension) and in number of lines (vertical dimension).

Figure 78. Coordinates and size of the window after cropping

These registers specify the coordinates of the starting point of the capture window as a line number (in the frame, starting from 0) and a number of pixel clocks (on the line, starting from 0), and the size of the window as a line number and a number of pixel clocks. The CAPCNT value can only be a multiple of 4 (two least significant bits are forced to 0) to allow the correct transfer of data through the DMA.

If the VSYNC signal goes active before the number of lines is specified in the DCMI_CWSIZE register, then the capture stops and an IT_FRAME interrupt is generated when enabled.

Figure 79. Data capture waveforms

1. Here, the active state of DCMI_HSYNC and DCMI_VSYNC is 1.
2. DCMI_HSYNC and DCMI_VSYNC can change states at the same time.

15.5.6 JPEG format

To allow JPEG image reception, it is necessary to set the JPEG bit in the DCMI_CR register. JPEG images are not stored as lines and frames, so the VSYNC signal is used to start the capture while HSYNC serves as a data enable signal. The number of bytes in a line may not be a multiple of 4, you should therefore be careful when handling this case since a DMA request is generated each time a complete 32-bit word has been constructed from the captured data. When an end of frame is detected and the 32-bit word to be transferred has not been completely received, the remaining data are padded with '0s' and a DMA request is generated.

The crop feature and embedded synchronization codes cannot be used in the JPEG format.

15.5.7 FIFO

A four-word FIFO is implemented to manage data rate transfers on the AHB. The DCMI features a simple FIFO controller with a read pointer incremented each time the camera interface reads from the AHB, and a write pointer incremented each time the camera interface writes to the FIFO. There is no overrun protection to prevent the data from being overwritten if the AHB interface does not sustain the data transfer rate.

In case of overrun or errors in the synchronization signals, the FIFO is reset and the DCMI interface waits for a new start of frame.

15.6 Data format description

15.6.1 Data formats

Three types of data are supported:

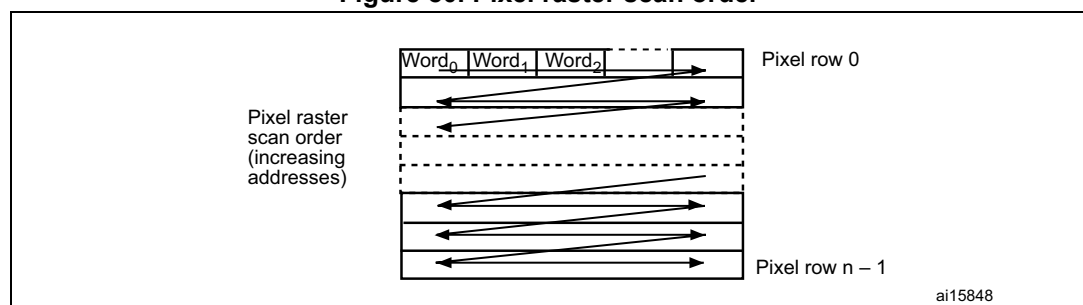
- 8-bit progressive video: either monochrome or raw Bayer format
- YCbCr 4:2:2 progressive video
- RGB565 progressive video. A pixel coded in 16 bits (5 bits for blue, 5 bits for red, 6 bits for green) takes two clock cycles to be transferred.

Compressed data: JPEG

For B&W, YCbCr or RGB data, the maximum input size is 2048×2048 pixels. No limit in JPEG compressed mode.

For monochrome, RGB & YCbCr, the frame buffer is stored in raster mode. 32-bit words are used. Only the little endian format is supported.

Figure 80. Pixel raster scan order



15.6.2 Monochrome format

Characteristics:

- Raster format
- 8 bits per pixel

[Table 83](#) shows how the data are stored.

Table 83. Data storage in monochrome progressive video format

Byte address	31:24	23:16	15:8	7:0
0	n + 3	n + 2	n + 1	n
4	n + 7	n + 6	n + 5	n + 4

15.6.3 RGB format

Characteristics:

- Raster format
- RGB
- Interleaved: one buffer: R, G & B interleaved: BRGBRBRG, etc.
- Optimized for display output

The RGB planar format is compatible with standard OS frame buffer display formats.

Only 16 BPP (bits per pixel): RGB565 (2 pixels per 32-bit word) is supported.

The 24 BPP (palletized format) and grayscale formats are not supported. Pixels are stored in a raster scan order, that is from top to bottom for pixel rows, and from left to right within a pixel row. Pixel components are R (red), G (green) and B (blue). All components have the same spatial resolution (4:4:4 format). A frame is stored in a single part, with the components interleaved on a pixel basis.

[Table 84](#) shows how the data are stored.

Table 84. Data storage in RGB progressive video format

Byte address	31:27	26:21	20:16	15:11	10:5	4:0
0	Red n + 1	Green n + 1	Blue n + 1	Red n	Green n	Blue n
4	Red n + 4	Green n + 3	Blue n + 3	Red n + 2	Green n + 2	Blue n + 2

15.6.4 YCbCr format

Characteristics:

- Raster format
- YCbCr 4:2:2
- Interleaved: one Buffer: Y, Cb & Cr interleaved: CbYCrYCbYCr, etc.

Pixel components are Y (luminance or “luma”), Cb and Cr (chrominance or “chroma” blue and red). Each component is encoded in 8 bits. Luma and chroma are stored together (interleaved) as shown in [Table 85](#).

Table 85. Data storage in YCbCr progressive video format

Byte address	31:24	23:16	15:8	7:0
0	Y _{n+1}	Cr _n	Y _n	Cb _n
4	Y _{n+3}	Cr _{n+2}	Y _{n+2}	Cb _{n+2}

15.7 DCMI interrupts

Five interrupts are generated. All interrupts are maskable by software. The global interrupt (IT_DCMI) is the OR of all the individual interrupts. [Table 86](#) gives the list of all interrupts.

Table 86. DCMI interrupts

Interrupt name	Interrupt event
IT_LINE	Indicates the end of line
IT_FRAME	Indicates the end of frame capture
IT_OVR	indicates the overrun of data reception
IT_VSYNC	Indicates the synchronization frame
IT_ERR	Indicates the detection of an error in the embedded synchronization frame detection
IT_DCMI	Logic OR of the previous interrupts

15.8 DCMI register description

All DCMI registers have to be accessed as 32-bit words, otherwise a bus error occurs.

15.8.1 DCMI control register 1 (DCMI_CR)

Address offset: 0x00

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
Reserved																	ENABLE	Reserved	EDM		FCRC		VSPOL	HSPOL	PCKPOL	ESS	JPEG	CROP	CM	CAPTURE			
																			RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW

Bit 31:15 Reserved, must be kept at reset value.

Bit 14 **ENABLE**: DCMI enable

0: DCMI disabled

1: DCMI enabled

Note: The DCMI configuration registers should be programmed correctly before enabling this Bit

Bit 13: 12 Reserved, must be kept at reset value.

Bits 11:10 **EDM[1:0]**: Extended data mode

- 00: Interface captures 8-bit data on every pixel clock
- 01: Interface captures 10-bit data on every pixel clock
- 10: Interface captures 12-bit data on every pixel clock
- 11: Interface captures 14-bit data on every pixel clock

Bits 9:8 **FCRC[1:0]**: Frame capture rate control

These bits define the frequency of frame capture. They are meaningful only in Continuous grab mode. They are ignored in snapshot mode.

- 00: All frames are captured
- 01: Every alternate frame captured (50% bandwidth reduction)
- 10: One frame in 4 frames captured (75% bandwidth reduction)
- 11: reserved

Bit 7 **VSPOL**: Vertical synchronization polarity

This bit indicates the level on the VSYNC pin when the data are not valid on the parallel interface.

- 0: VSYNC active low
- 1: VSYNC active high

Bit 6 **HSPOL**: Horizontal synchronization polarity

This bit indicates the level on the HSYNC pin when the data are not valid on the parallel interface.

- 0: HSYNC active low
- 1: HSYNC active high

Bit 5 **PCKPOL**: Pixel clock polarity

This bit configures the capture edge of the pixel clock

- 0: Falling edge active.
- 1: Rising edge active.

Bit 4 **ESS**: Embedded synchronization select

- 0: Hardware synchronization data capture (frame/line start/stop) is synchronized with the HSYNC/VSYNC signals.
- 1: Embedded synchronization data capture is synchronized with synchronization codes embedded in the data flow.

Note: Valid only for 8-bit parallel data. HSPOL/VSPOL are ignored when the ESS bit is set.

This bit is disabled in JPEG mode.

Bit 3 **JPEG**: JPEG format

- 0: Uncompressed video format
- 1: This bit is used for JPEG data transfers. The HSYNC signal is used as data enable. The crop and embedded synchronization features (ESS bit) cannot be used in this mode.

Bits 2 **CROP**: Crop feature

- 0: The full image is captured. In this case the total number of bytes in an image frame should be a multiple of 4
- 1: Only the data inside the window specified by the crop register are captured. If the size of the crop window exceeds the picture size, then only the picture size is captured.

Bit 1 **CM**: Capture mode

0: Continuous grab mode - The received data are transferred into the destination memory through the DMA. The buffer location and mode (linear or circular buffer) is controlled through the system DMA.

1: Snapshot mode (single frame) - Once activated, the interface waits for the start of frame and then transfers a single frame through the DMA. At the end of the frame, the CAPTURE bit is automatically reset.

Bit 0 **CAPTURE**: Capture enable

0: Capture disabled.

1: Capture enabled.

The camera interface waits for the first start of frame, then a DMA request is generated to transfer the received data into the destination memory.

In snapshot mode, the CAPTURE bit is automatically cleared at the end of the 1st frame received.

In continuous grab mode, if the software clears this bit while a capture is ongoing, the bit is effectively cleared after the frame end.

Note: The DMA controller and all DCMI configuration registers should be programmed correctly before enabling this bit.

15.8.2 DCMI status register (DCMI_SR)

Address offset: 0x04

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved																														FNE	VSYNC	HSYNC
																														r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **FNE**: FIFO not empty

This bit gives the status of the FIFO

1: FIFO contains valid data

0: FIFO empty

Bit 1 **VSYNC**

This bit gives the state of the VSYNC pin with the correct programmed polarity. When embedded synchronization codes are used, the meaning of this bit is the following:

0: active frame

1: synchronization between frames

In case of embedded synchronization, this bit is meaningful only if the CAPTURE bit in DCMI_CR is set.

Bit 0 HSYNC

This bit gives the state of the HSYNC pin with the correct programmed polarity. When embedded synchronization codes are used, the meaning of this bit is the following:

0: active line

1: synchronization between lines

In case of embedded synchronization, this bit is meaningful only if the CAPTURE bit in DCMI_CR is set.

15.8.3 DCMI raw interrupt status register (DCMI_RIS)

Address offset: 0x08

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved																												LINE_RIS	VSYNC_RIS	ERR_RIS	OVR_RIS	FRAME_RIS
																												1	1	1	1	1

DCMI_RIS gives the raw interrupt status and is accessible in read only. When read, this register returns the status of the corresponding interrupt before masking with the DCMI_IER register value.

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 LINE_RIS: Line raw interrupt status

This bit gets set when the HSYNC signal changes from the inactive state to the active state. It goes high even if the line is not valid.

In the case of embedded synchronization, this bit is set only if the CAPTURE bit in DCMI_CR is set.

It is cleared by writing a '1' to the LINE_ISC bit in DCMI_ICR.

Bit 3 VSYNC_RIS: VSYNC raw interrupt status

This bit is set when the VSYNC signal changes from the inactive state to the active state.

In the case of embedded synchronization, this bit is set only if the CAPTURE bit is set in DCMI_CR.

It is cleared by writing a '1' to the VSYNC_ISC bit in DCMI_ICR.

Bit 2 ERR_RIS: Synchronization error raw interrupt status

0: No synchronization error detected

1: Embedded synchronization characters are not received in the correct order.

This bit is valid only in the embedded synchronization mode. It is cleared by writing a '1' to the ERR_ISC bit in DCMI_ICR.

Note: This bit is available only in embedded synchronization mode.

Bit 1 OVR_RIS: Overrun raw interrupt status

0: No data buffer overrun occurred

1: A data buffer overrun occurred and the data FIFO is corrupted.

This bit is cleared by writing a '1' to the OVR_ISC bit in DCMI_ICR.

Bit 0 **FRAME_RIS**: Capture complete raw interrupt status

0: No new capture

1: A frame has been captured.

This bit is set when a frame or window has been captured.

In case of a cropped window, this bit is set at the end of line of the last line in the crop. It is set even if the captured frame is empty (e.g. window cropped outside the frame).

This bit is cleared by writing a '1' to the FRAME_ISC bit in DCMI_ICR.

15.8.4 DCMI interrupt enable register (DCMI_IER)

Address offset: 0x0C

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																										LINE_IE	VSYNC_IE	ERR_IE	OVR_IE	FRAME_IE	
																										rw	rw	rw	rw	rw	

The DCMI_IER register is used to enable interrupts. When one of the DCMI_IER bits is set, the corresponding interrupt is enabled. This register is accessible in both read and write.

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 **LINE_IE**: Line interrupt enable

0: No interrupt generation when the line is received

1: An Interrupt is generated when a line has been completely received

Bit 3 **VSYNC_IE**: VSYNC interrupt enable

0: No interrupt generation

1: An interrupt is generated on each VSYNC transition from the inactive to the active state

The active state of the VSYNC signal is defined by the VSPOL bit.

Bit 2 **ERR_IE**: Synchronization error interrupt enable

0: No interrupt generation

1: An interrupt is generated if the embedded synchronization codes are not received in the correct order.

Note: This bit is available only in embedded synchronization mode.

Bit 1 **OVR_IE**: Overrun interrupt enable

0: No interrupt generation

1: An interrupt is generated if the DMA was not able to transfer the last data before new data (32-bit) are received.

Bit 0 **FRAME_IE**: Capture complete interrupt enable

0: No interrupt generation

1: An interrupt is generated at the end of each received frame/crop window (in crop mode).

15.8.5 DCMI masked interrupt status register (DCMI_MIS)

This DCMI_MIS register is a read-only register. When read, it returns the current masked status value (depending on the value in DCMI_IER) of the corresponding interrupt. A bit in this register is set if the corresponding enable bit in DCMI_IER is set and the corresponding bit in DCMI_RIS is set.

Address offset: 0x10

Reset value: 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved																												LINE_MIS	VSYNC_MIS	ERR_MIS	OVR_MIS	FRAME_MIS
																												r	r	r	r	r

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 **LINE_MIS**: Line masked interrupt status

This bit gives the status of the masked line interrupt

0: No interrupt generation when the line is received

1: An Interrupt is generated when a line has been completely received and the LINE_IE bit is set in DCMI_IER.

Bit 3 **VSYNC_MIS**: VSYNC masked interrupt status

This bit gives the status of the masked VSYNC interrupt

0: No interrupt is generated on VSYNC transitions

1: An interrupt is generated on each VSYNC transition from the inactive to the active state and the VSYNC_IE bit is set in DCMI_IER.

The active state of the VSYNC signal is defined by the VSPOL bit.

Bit 2 **ERR_MIS**: Synchronization error masked interrupt status

This bit gives the status of the masked synchronization error interrupt

0: No interrupt is generated on a synchronization error

1: An interrupt is generated if the embedded synchronization codes are not received in the correct order and the ERR_IE bit in DCMI_IER is set.

Note: This bit is available only in embedded synchronization mode.

Bit 1 **OVR_MIS**: Overrun masked interrupt status

This bit gives the status of the masked overflow interrupt

0: No interrupt is generated on overrun

1: An interrupt is generated if the DMA was not able to transfer the last data before new data (32-bit) are received and the OVR_IE bit is set in DCMI_IER.

Bit 0 **FRAME_MIS**: Capture complete masked interrupt status

This bit gives the status of the masked capture complete interrupt

0: No interrupt is generated after a complete capture

1: An interrupt is generated at the end of each received frame/crop window (in crop mode) and the FRAME_IE bit is set in DCMI_IER.

15.8.6 DCMI interrupt clear register (DCMI_ICR)

Address offset: 0x14

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																										LINE_ISC	VSYNC_ISC	ERR_ISC	OVR_ISC	FRAME_ISC	
																										W	W	W	W	W	

The DCMI_ICR register is write-only. Writing a '1' into a bit of this register clears the corresponding bit in the DCMI_RIS and DCMI_MIS registers. Writing a '0' has no effect.

Bits 15:5 Reserved, must be kept at reset value.

Bit 4 **LINE_ISC**: line interrupt status clear

Writing a '1' into this bit clears LINE_RIS in the DCMI_RIS register

Bit 3 **VSYNC_ISC**: Vertical synch interrupt status clear

Writing a '1' into this bit clears the VSYNC_RIS bit in DCMI_RIS

Bit 2 **ERR_ISC**: Synchronization error interrupt status clear

Writing a '1' into this bit clears the ERR_RIS bit in DCMI_RIS

Note: This bit is available only in embedded synchronization mode.

Bit 1 **OVR_ISC**: Overrun interrupt status clear

Writing a '1' into this bit clears the OVR_RIS bit in DCMI_RIS

Bits 0 **FRAME_ISC**: Capture complete interrupt status clear

Writing a '1' into this bit clears the FRAME_RIS bit in DCMI_RIS

15.8.7 DCMI embedded synchronization code register (DCMI_ESCR)

Address offset: 0x18

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FEC								LEC								LSC								FSC							
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:24 FEC: Frame end delimiter code

This byte specifies the code of the frame end delimiter. The code consists of 4 bytes in the form of 0xFF, 0x00, 0x00, FEC.

If FEC is programmed to 0xFF, all the unused codes (0xFF0000XY) are interpreted as frame end delimiters.

Bits 23:16 LEC: Line end delimiter code

This byte specifies the code of the line end delimiter. The code consists of 4 bytes in the form of 0xFF, 0x00, 0x00, LEC.

Bits 15:8 LSC: Line start delimiter code

This byte specifies the code of the line start delimiter. The code consists of 4 bytes in the form of 0xFF, 0x00, 0x00, LSC.

Bits 7:0 FSC: Frame start delimiter code

This byte specifies the code of the frame start delimiter. The code consists of 4 bytes in the form of 0xFF, 0x00, 0x00, FSC.

If FSC is programmed to 0xFF, no frame start delimiter is detected. But, the 1st occurrence of LSC after an FEC code is interpreted as a start of frame delimiter.

15.8.8 DCMI embedded synchronization unmask register (DCMI_ESUR)

Address offset: 0x1C

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FEU								LEU								LSU								FSU							
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:24 FEU: Frame end delimiter unmask

This byte specifies the mask to be applied to the code of the frame end delimiter.

0: The corresponding bit in the FEC byte in DCMI_ESCR is masked while comparing the frame end delimiter with the received data.

1: The corresponding bit in the FEC byte in DCMI_ESCR is compared while comparing the frame end delimiter with the received data

Bits 23:16 LEU: Line end delimiter unmask

This byte specifies the mask to be applied to the code of the line end delimiter.

0: The corresponding bit in the LEC byte in DCMI_ESCR is masked while comparing the line end delimiter with the received data

1: The corresponding bit in the LEC byte in DCMI_ESCR is compared while comparing the line end delimiter with the received data

Bits 15:8 LSU: Line start delimiter unmask

This byte specifies the mask to be applied to the code of the line start delimiter.

0: The corresponding bit in the LSC byte in DCMI_ESCR is masked while comparing the line start delimiter with the received data

1: The corresponding bit in the LSC byte in DCMI_ESCR is compared while comparing the line start delimiter with the received data

Bits 7:0 **FSU**: Frame start delimiter unmask

This byte specifies the mask to be applied to the code of the frame start delimiter.

0: The corresponding bit in the FSC byte in DCMI_ESCR is masked while comparing the frame start delimiter with the received data

1: The corresponding bit in the FSC byte in DCMI_ESCR is compared while comparing the frame start delimiter with the received data

15.8.9 DCMI crop window start (DCMI_CWSTRT)

Address offset: 0x20

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			VST[12:0]														Reserved		HOFFCNT[13:0]												
			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bits 28:16 **VST[12:0]**: Vertical start line count

The image capture starts with this line number. Previous line data are ignored.

0x0000 => line 1

0x0001 => line 2

0x0002 => line 3

....

Bits 15:14 Reserved, must be kept at reset value.

Bit 13:0 **HOFFCNT[13:0]**: Horizontal offset count

This value gives the number of pixel clocks to count before starting a capture.

15.8.10 DCMI crop window size (DCMI_CWSIZE)

Address offset: 0x24

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			VLIN[13:0]														Reserved		CAPCNT[13:0]												
			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:16 **VLIN[13:0]**: Vertical line count

This value gives the number of lines to be captured from the starting point.

0x0000 => 1 line

0x0001 => 2 lines

0x0002 => 3 lines

....

Bits 15:14 Reserved, must be kept at reset value.

Bits 13:0 **CAPCNT[13:0]**: Capture count

This value gives the number of pixel clocks to be captured from the starting point on the same line. Its value should correspond to word-aligned data for different widths of parallel interfaces.

0x0000 => 1 pixel

0x0001 => 2 pixels

0x0002 => 3 pixels

....

15.8.11 DCMI data register (DCMI_DR)

Address offset: 0x28

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte3								Byte2								Byte1								Byte0							
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:24 Data byte 3

Bits 23:16 Data byte 2

Bits 15:8 Data byte 1

Bits 7:0 Data byte 0

The digital camera Interface packages all the received data in 32-bit format before requesting a DMA transfer. A 4-word deep FIFO is available to leave enough time for DMA transfers and avoid DMA overrun conditions.

15.8.12 DCMI register map

[Table 87](#) summarizes the DCMI registers.

Table 87. DCMI register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	DCMI_CR	Reserved																	ENABLE	Reserved	EDM		FCR		VSPOL	HSPOL	PCKPOL	ESS	JPEG	CROP	CM	CAPTURE	
	Reset value																		0														0
0x04	DCMI_SR	Reserved																									FNE			VSYNC	HSYNC		
	Reset value																													0	0	0	
0x08	DCMI_RIS	Reserved																									LINE_RIS	VSYNC_RIS	ERR_RIS	OVR_RIS	FRAME_RIS		
	Reset value																										0	0	0	0	0		

Table 87. DCMI register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0C	DCMI_IER	Reserved																										LINE_IE	VSYNC_IE	ERR_IE	OVR_IE	FRAME_IE	
	Reset value																											0	0	0	0	0	
0x10	DCMI_MIS	Reserved																										LINE_MIS	VSYNC_MIS	ERR_MIS	OVR_MIS	FRAME_MIS	
	Reset value																											0	0	0	0	0	
0x14	DCMI_ICR	Reserved																										LINE_ISC	VSYNC_ISC	ERR_ISC	OVR_ISC	FRAME_ISC	
	Reset value																											0	0	0	0	0	
0x18	DCMI_ESCR	FEC								LEC								LSC								FSC							
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x1C	DCMI_ESUR	FEU								LEU								LSU								FSU							
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x20	DCMI_CWSTR_T	Reserved	VST[12:0]												Reserved	HOFFCNT[13:0]																	
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x24	DCMI_CWSIZE	Reserved	VLINE[13:0]												Reserved	CAPCNT[13:0]																	
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x28	DCMI_DR	Byte3								Byte2								Byte1								Byte0							
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.